

Harvard (HEEDB) EEG report labels — the fields and how they were extracted

The Harvard EEG Database ships a per-recording label table (*_EEG_with_reports) with 39 finding columns plus identifier/metadata columns. This note lists every field, what it means, and how it was produced. As of now this is **verified**, not reconstructed: the data-descriptor paper’s supplement (Sun et al., Epilepsia 2025, **Data S1**) gives the extraction method, the prompt, and the keyword lists.

How the labels were made — two channels

Data S1 §A describes two separate extraction pipelines over two data sources:

- **From EEG annotations → keyword + regular-expression matching** (Data S1 Table S5). This is the plain keyword search.
- **From EEG reports → a fine-tuned LLM** (a Q&A fine-tuned Llama 3, “Bio-Medical-Llama-3-8B” on Hugging Face). Each finding is one templated question, with a **three-way** answer:

“Based on the medical context of neurology and EEG analysis, does the following report indicate the presence of **[finding]** in the EEG? Please respond with **yes, no, or not mentioned.**”

The labels we compared against are the **report / LLM channel** (*_EEG_with_reports), not the keyword channel — so our MedGemma-vs-Bio-Medical-Llama comparison is model-vs-model, as intended. Two more points from Data S1 §A.4 that matter for reading the columns:

- An empty finding column means the model answered **not mentioned**, which is *not* the same as “absent”; the paper only measures agreement on the “yes” cells (so present = column non-empty is the right reading, as we used).
- The report labels are “reviewed and confirmed by at least two clinicians, including a senior clinician,” so they are treated as relatively stronger than the annotation labels.

The 39 finding fields

Meaning is the standard-EEG reading of the column name. The keyword column is Harvard’s actual Table S5 regex for the **annotation** channel (the report channel used the LLM prompt above, not these keywords) — it is shown because it is their real, published definition and it reveals how narrow some fields are.

#	Field	What it means (standard EEG terminology)	Annotation keyword (Data S1 Table S5)
1	normal	EEG reported normal / within normal limits	normal
2	abnormal	EEG reported abnormal (any abnormal finding)	abnormal
3	awake	Awake state captured	eo, awake
4	n1	Sleep stage N1	drowsiness, drowsy, n1
5	n2	Sleep stage N2	spindle, spindles, ss, kss, n2, ii
6	bets	Benign epileptiform transients of sleep (small sharp spikes)	bets, sss
7	wicket	Wicket spikes (benign variant)	wicket[s]
8	spindles	Sleep spindles	spindle, spindles, ss, kss
9	vertex wave	Vertex sharp transients	v, vw, vert, vertw, vertwave, vertex, vertexwave

#	Field	What it means (standard EEG terminology)	Annotation keyword (Data S1 Table S5)
10	posts	Positive occipital sharp transients of sleep	posts
11	breach	Breach rhythm (activity over a skull defect)	bre[ea]ch
12	spikes	Epileptiform spikes / sharp waves	spike[s], discharge[s], sharp
13	seizure	Electrographic / clinical seizure	sz, serizure[s]
14	lpd	Lateralized periodic discharges	lp[e]d[s], pled
15	gpd	Generalized periodic discharges	gpd[s]
16	lrda	Lateralized rhythmic delta activity	lrda, tirda
17	grda	Generalized rhythmic delta activity	grda, [of]irda
18	bs	Burst-suppression	burst, suppression, supp, bsr
19	foc slowing	Focal slowing	foc* slow*, r* slow*, l* slow*, l* temporal slow*
20	gen slowing	Generalized / diffuse slowing	gen* slow*
21	pdr	Posterior dominant rhythm (normal background)	pdr
22	dravet	Dravet syndrome pattern	<i>(not in Data S1 — undocumented)</i>
23	eses	Electrical status epilepticus in sleep	eses
24	fold	Unclear — appears in Data S1 figures but has no keyword or definition	<i>(none)</i>
25	jae	Juvenile absence epilepsy	jea
26	jeavons	Jeavons syndrome	<i>(not in Data S1 — undocumented)</i>
27	jme	Juvenile myoclonic epilepsy	hme
28	k_complexes	K-complexes	k-complexes, k
29	bects	Benign epilepsy with centrotemporal spikes	bects
30	bipd	Bilateral independent periodic discharges	bipd
31	cjd	Creutzfeldt-Jakob periodic pattern	cjd
32	diffuse Beta	Diffuse / excessive beta	diffuse beta
33	low voltage	Low-voltage background	low voltage
34	ppr	Photoparoxysmal response	ppr
35	status	Status epilepticus	<i>(listed as a finding, no keyword given)</i>
36	sunflower	Sunflower syndrome	<i>(not in Data S1 — undocumented)</i>
37	uninterpretable	Recording uninterpretable	<i>(in figures, no keyword)</i>

#	Field	What it means (standard EEG terminology)	Annotation keyword (Data S1 Table S5)
38	wham	Unclear — not standard EEG terminology, absent from Data S1	(none)
39	angelman	Angelman syndrome pattern	angelman

Two nuances worth knowing: the keyword lists carry small typos (seizure serizure, JAE jea, JME hme) — but those only affect the annotation channel, not our report labels. And six DB columns (dravet, fold, jeavons, sunflower, uninterpretable, wham) fall outside the paper’s 33 documented findings, so they are effectively undocumented.

The identifier / metadata columns

SiteID, BDSPPatientID, SessionID, AgeAtVisit, SexDSC, ServiceName(EEG), CreationTime(EEG), StartTime(EEG), EndTime(EEG), ProcedureDSC(Reports), EncounterDTS(Reports), BeginDTS(Reports), ExamEndDTS(Reports), DeidentifiedName(Reports).

Who read each report

There is no author column. The reading physician appears only inside the report text, as a signature — e.g. *“I (... , MD) have reviewed the study and edited this report”* (sometimes “...with the fellow”, meaning a fellow drafted and the attending signed off). Data S1 §D explains the masking: reports are de-identified with **PHILter** (rule-based + statistical PHI removal, HIPAA Safe Harbor, with date shifting), which is why most signatures collapse to ***** while a few readers’ initials slip through. Coverage of that signature line, by hospital:

Hospital	Reports	Has “I (...)” signature	Any attestation phrase*	Distinguishable reader**
MGH	55,557	35,652 (64%)	54,233 (98%)	8,182
BWH	21,035	2 (0%)	11,619 (55%)	2
BCH	25,022	0 (0%)	7,297 (29%)	0
Total	101,614	35,654 (35%)	73,149 (72%)	8,184 (8%)

* “edited this report / reviewed the study / personally reviewed / electronically signed” etc. ** signature keeps initials, so individual readers can be told apart; the rest collapse to a masked *****. The distinguishable readers (≈14, almost all MGH: ADL, ESR, LM, DBH, JS, CMM, LTG, KCS, ...) are listed in results/harvard/report_authors.csv.

How this bears on our comparison

Only a few of these map 1:1 to one of our five categories. abnormal is the clean one, which is why the comparison report keeps to it. Table S5 confirms why the slowing findings do not line up: Harvard’s foc slowing keyword is foc* slow*, r* slow*, l* slow* and gen slowing is gen* slow* — i.e. **only the word “slowing”** (with laterality) — whereas our focal / generalized non-epileptiform categories bundle slowing together with attenuation, asymmetry, disorganization, and excessive beta. Their own definitions, not just our empirical check (analysis/mapping_check.py), show the mismatch.

Reliability, per Data S1 §A.4

There is no gold standard, so they report an “ambiguity” rate — the share of true-yes cases the LLM missed (called no): **2.25%** on average against annotations, and **0%** against multi-expert labels (the LLM never said no where experts said yes). So the report LLM is high-recall and rarely denies a finding that is present.

Sources: Sun et al., *Harvard Electroencephalography Database*, *Epilepsia* 2025 (doi 10.1111/epi.18487), **Data S1** supplement (§A extraction method + prompt, Table S5 keywords, Table S2 finding counts); label schema — *_EEG_with_reports in HarvardEEG.db; reader signatures — results/harvard/report_authors.csv.